

ChronoFork🕒🔗:

Learner—Multi-Agent Co-Roleplay for Divergent Historical Experiences

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app.chronofork.me

<https://github.com/IS492-SP26/team-project-chronofork-history-rewrite-simulation>



1. Recap & Final Goals



Who we're building for

- History learners
- Engaging counterfactual reasoning exp
- Not formal classrooms



ChatGPT / Claude — great at subtasks

- But can't create a learning EXPERIENCE

MOST REWARDING FEEDBACK

***“It turned me from a reader
into an author.”***

— P4, post-study interview

BACKED BY THE NUMBERS

+3.11 on Sense of Agency (PENS, 1–7)

6 / 6 participants moved in the same direction

2. Evaluation Design Overview

Participants

- N = 6 (P1–P6, university students 18–25)
- Within-subject, counter-balanced order
- Each chose 2 events from the Event Library; 1 per system
- Compensation: \$10 / hr
- Recruited via online groups

Two Conditions

- ChronoFork-Web (dynamic)
- IDN-Twine (static)
- Both canonical, represented by ChronoFork-Web expert-audit
- Lab session

Metrics Collected

- NES (12 items) — Narrative Engagement
- ESS (7 items) — Explanation Satisfaction
- PENS Autonomy (3 + 2 items) — Agency
- UMUX-Lite (2) + Paas Mental Effort (1)
- IMI — Interest, Value, Pressure
- Semi-structured interview (~20 min)

3. Study Materials & Procedure



1. Intro & Consent

~5 min

briefing, sign consent,
event selection

2. Task — Exp. A

~17 min

tutorial video → 15 min
think-aloud

3. Survey + Break

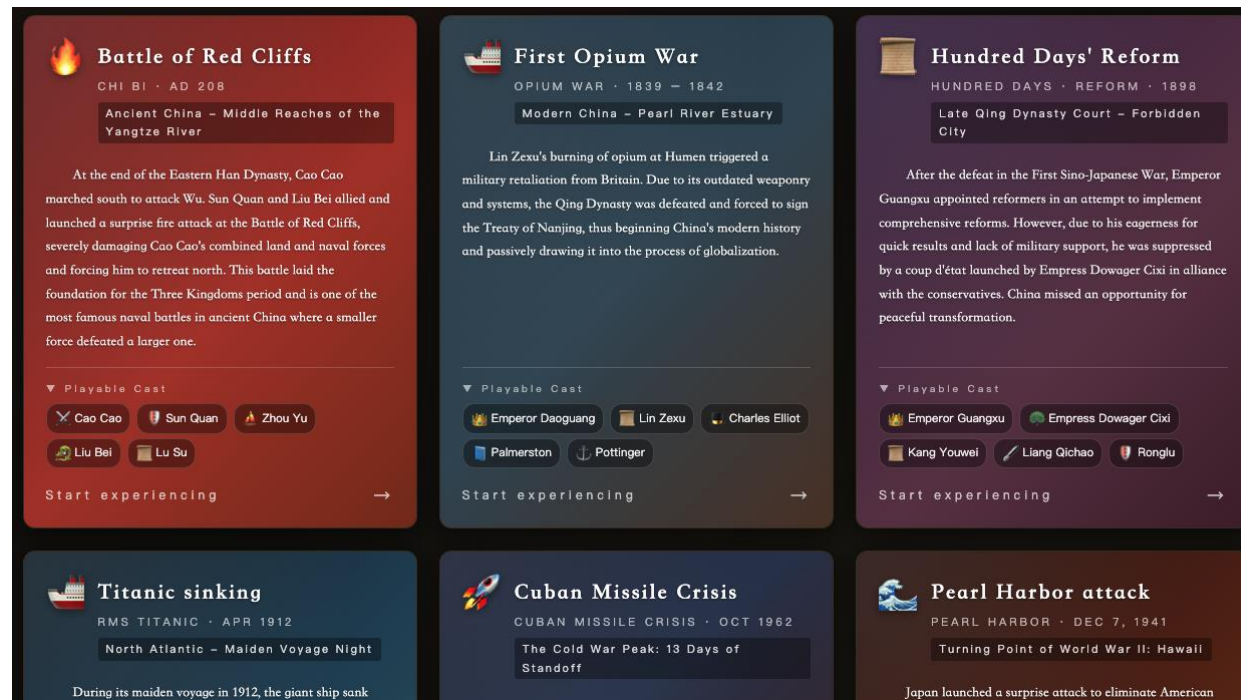
~7 min

Q1 (~5 min) + 2-min rest

4. Task B + Q2 + Interview

~30 min

mirror Exp. A, then ~20 min
interview



3. Study Materials & Procedure





Titanic sinking

RMS TITANIC · APRIL 14–15, 1912 · NORTH ATLANTIC



Titanic sinking

NORTH ATLANTIC · 1 9 1 2

In April 1912, the British passenger liner Titanic, touted as "unsinkable," set sail from Southampton, bound for New York. On board were 2,224 people—wealthy first-class passengers, middle-class second-class passengers, immigrants in the lower decks, and the entire crew. The safety of the entire voyage depended on the precise execution of every order from Captain Smith on the bridge, and the subtle, almost imperceptible, "expectation of speed" exerted behind the scenes by White Star Line general manager Ismay. On the Atlantic route, iceberg warnings were arriving one after another from other ships—a disaster, repeatedly examined by posterity, was quietly tightening the thread of fate for over 1,500 lives.

[Begin this history →](#)

Characters CAST

**Edward John Smith**
Captain of the Titanic
The highest-ranking officer on board is responsible for decisions regarding the course, speed, and overall safety.

**Joseph Bruce Ismay**
General Manager of White Star Line
He personally oversaw the maiden voyage and was seen as supporting the high speeds required to achieve a record-breaking arrival.

**Thomas Andrews**
Titanic's chief designer
Those who accompany the ship on inspections are the most knowledgeable about the ship's buoyancy.

**Henry Wilde**
Chief Mate
To carry out the captain's orders and directly command the deployment of lifeboats and the order of boarding.

**Arthur Rostrom**
Captain of the Carpathia
Upon receiving the distress call, the decision was made to sail at full speed through the night into the icy area to carry out the rescue.

Baseline (IDN)
(twine.chronofork.me)

Act One: The Battle of the Bridge

Luxury maiden voyage and speed pressure

SCENE 01

On the afternoon of April 14, the Titanic had received seven ice condition telegrams, forecasting a large amount of floating ice and icebergs between 41 and 42 degrees north latitude. The sea was calm and the visibility was excellent, but there would be no moon that night. Ismay, in casual conversation with Smith on the bridge, often mentioned that "if we could arrive in port half a day earlier than the Olympic, that would be headline news."

Smith, clutching the ice condition telegram, gazed at the sea under the setting sun. Should he reduce speed as usual for safety, or maintain the high speed of 22 knots and hope that "the lookout will spot it in time"? This soon-to-retire captain had to make the most important decision of his life, balancing honor and caution.

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Story Progress

- 1. **Luxury maiden voyage and speed pressure**
- 2. Discussion on the stillness of the deck and the drills
- 3. Iceberg Collision and the Chief's Judgment
- 4. Radio Room and Rescue Decisions
- 5. Rescued in the early morning and the question that has been asked for centuries

**Joseph Bruce Ismay**

"Captain, the Olympic took 6 days and 23 hours to sail west last time; we're confident we can beat her by half a day and night. There will be photographers waiting for us at the New York docks."

**Thomas Andrews**

"Captain, structurally and in terms of bulkheads, this ship can withstand reasonable sea conditions; however, that is, we must first be able to see the ice and then have time to get around it."

**Edward John Smith**

"I have received seven telegrams regarding the ice situation; the sea is calm but there is no moon—I must decide on the nighttime speed now."

▼ Key Decision

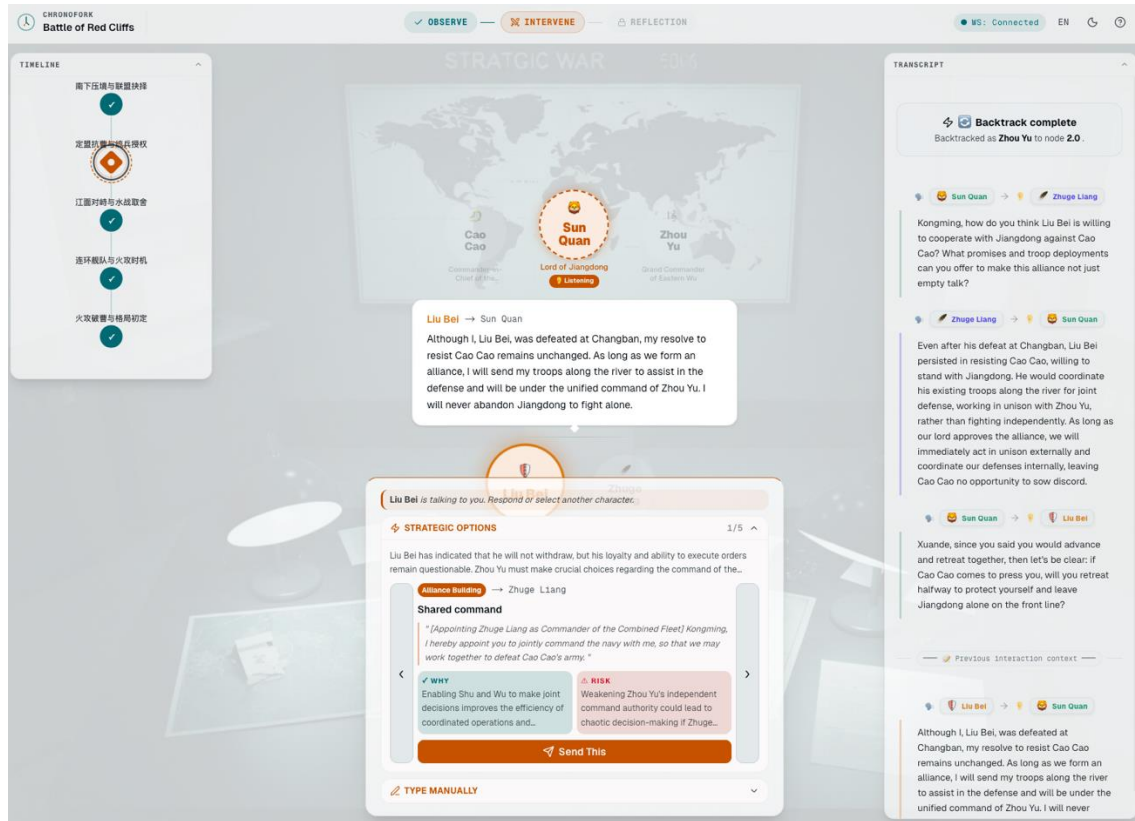
On the eve of the ice season, what decision should Smith make?

Decided by Edward John Smith

 **Reduce speed to a safe 12 knots and double the lookout.**
Conservative approach—a safe route deviating from official history

 **Maintain near full speed at 22 knots as you enter the ice zone**
The Pursuit of Records—Selection of Official History

3. Study Materials & Procedure



ChronoFork Web
(app.chronofork.me)

TIMELINE VERDICT · AMBIGUOUS

Battle of Red Cliffs
EPISODE COMPLETE

Alternative Endings & Mysteries

Alternative Endings & Mysteries · Deviates from official history

Fire attack launched first, resulting in a stalemate at Red Cliffs.

A single fire awakened Cao Cao, dragging the battle into a deep abyss of attrition.

The Story Path You've Followed

- The flames ignited first: the fire attack became a warning bell.**

Zhou Yu launched a surprise attack by setting fires ahead of time, attempting to disrupt the rhythm of Cao Cao's naval forces with a single blow.
- River blockade: crossbows and small boats woven into an iron net**

Cao Cao used patrol boats and crossbows to press down on the river, making it difficult for the allied forces to advance and harass the enemy again.
- Returning to the village and holding firm: Wind direction and intelligence replace high-stakes gambles**

Zhou Yu chose to return to the naval stronghold to reorganize and hold it, coordinating with Zhuge Liang on wind direction and intelligence, and waiting for a window of opportunity to control the situation.

Counterfactual deduction

In this rewritten narrative, the Battle of Red Cliffs is no longer a blazing myth of overnight victory, but a protracted stalemate dragged into a quagmire by a "prematurely exposed ingenious plan." Zhou Yu's caution prevented the collapse of the battle lines, but it also turned the decisive moment into a period of attrition. History ultimately records not who burned the fleet, but who lost patience and resources first in the stalemate.

Characters CAST

Cao Cao

Commander-in-Chief of the Northern Alliance

Sun Quan

Lord of Jiangdong

Zhou Yu

Grand Commander of Eastern Wu

Liu Bei

Jingzhou exiled leader

Zhugge Liang

Liu Bei's strategist

Story Progress

1. The initial flames: The fire attack became a warning bell.

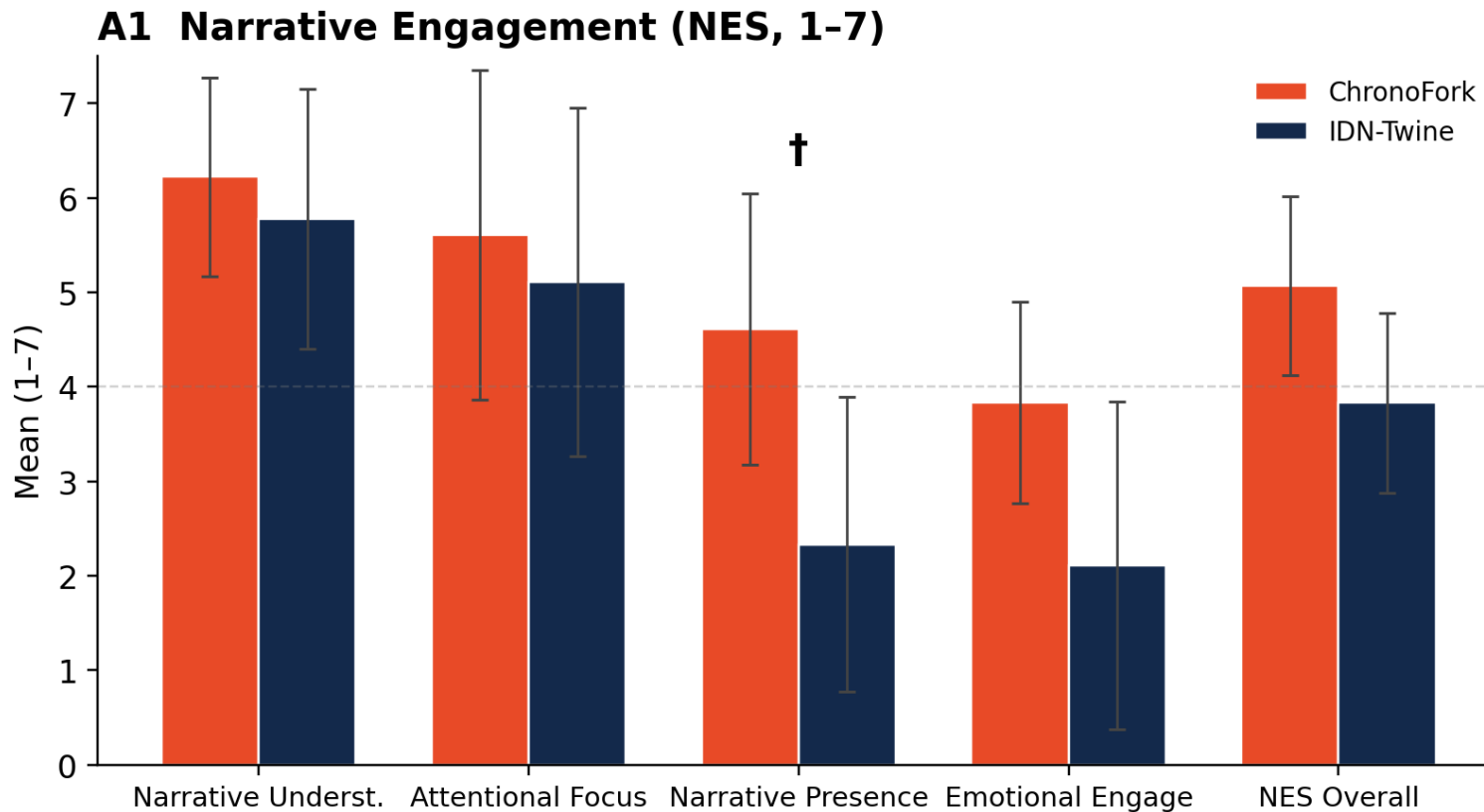
2. River blockade: Crossbows and small boats were woven into an iron net

3. Retreat to the village and hold firm: Wind direction and intelligence replace high-stakes gambles

4. Quantitative Results



Narrative Engagement (N = 6, Wilcoxon two-sided)



Narrative Presence +2.28

$p = .094$

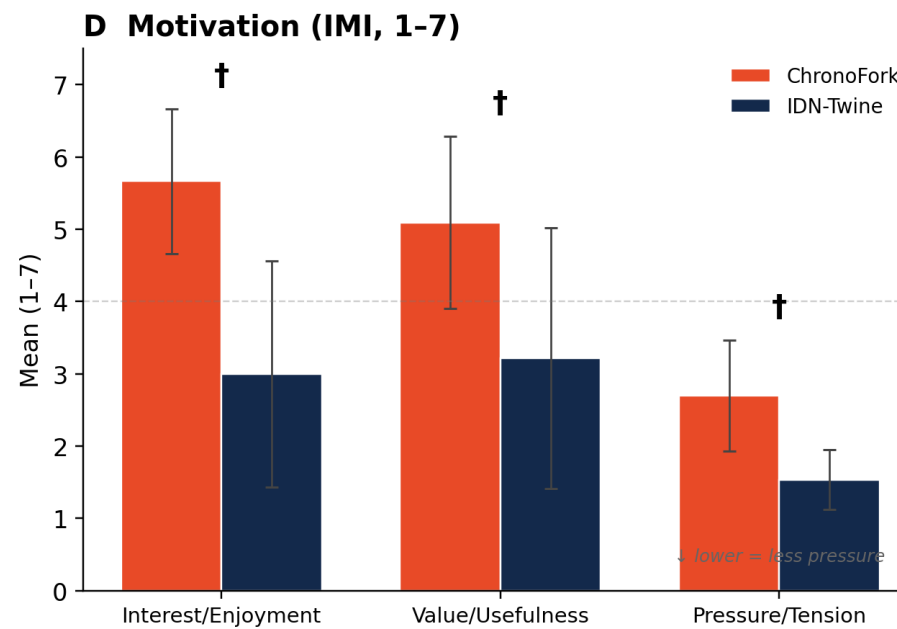
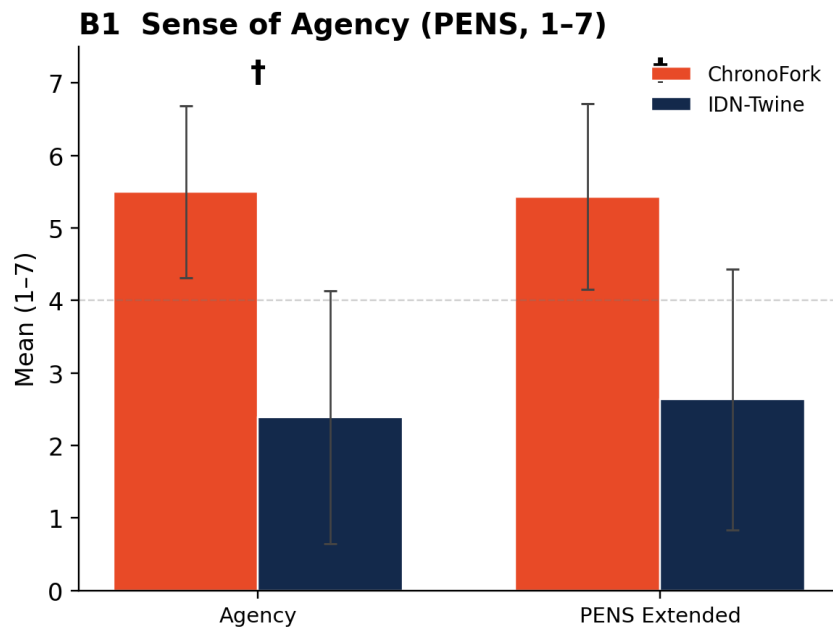
$† p < .10$

$N=6 \rightarrow$ smallest possible p (two-sided) is .063

4. Quantitative Results



Sense of Agency (N = 6, Wilcoxon two-sided)



Agency +3.11

$p = .063$ (6 / 6 same direction)

Interest / Enjoyment +2.67

$p = .063$

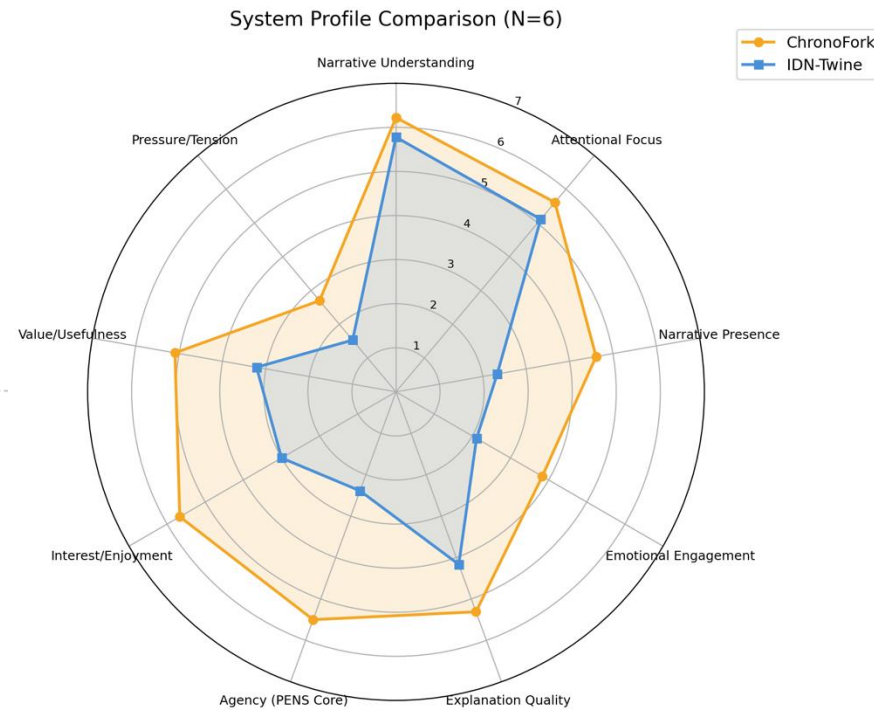
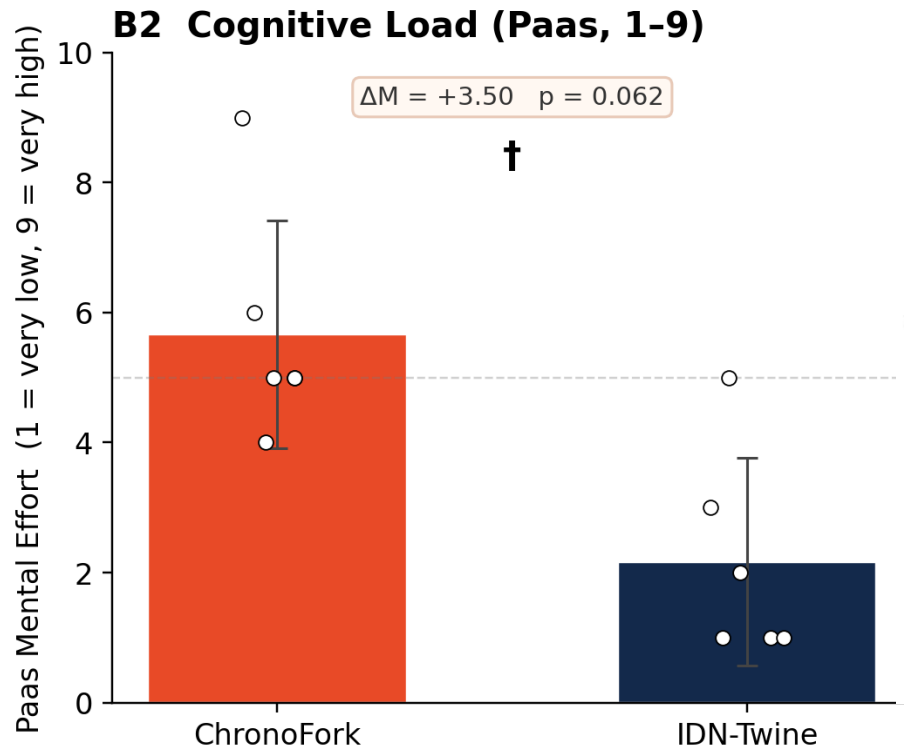
† $p < .10$

$N=6 \rightarrow$ smallest possible p (two-sided) is .063

4. Quantitative Results



Sense of Agency (N = 6, Wilcoxon two-sided)



Mental Effort +3.50 (1-9)

$p = .063$ higher = more effort

UMUX-Lite -0.08

(66.7 vs 68.1 / 100)

$p = 1.00$ honest tie

† $p < .10$

$N=6 \rightarrow$ smallest possible p (two-sided) is .063



5. Interpretation & Discussion

Orchestration delivered the experience

- Agency, Presence, Interest, Value — all 6 / 6 same direction
- Effect sizes large ($r \geq 0.77$) on every key narrative dimension
- "Reader → author" reframing reported by all 6 interviewees
- CP3 success criteria (engagement + critical thinking) supported relative to IDN baseline
- Confirms CP2's thesis: the missing layer was orchestration, not generation.

Higher mental effort may = germane, not extraneous

- Paas mental effort: ChronoFork 5.67 vs. Twine 2.17
- But UMUX-Lite is a tie → the load is not from the UI
- Pressure / Tension also slightly higher → active improvisation
- P2: "I had to think on my feet — that's why it felt alive."
- Reframe: the cost of being an author is doing the authoring.

6. Limitations, Risks & Ethics



Sample

- N = 6 pilot scale only
- Mandarin-speaking university students 18–25
- Convenience sample — campus posters & online groups
- Counter-balanced order, but practice effects possible

Methodology

- Open-ended exploration → no strict task-success metric
- 15-min cap per condition → may favor quick-to-read systems
- Lab setting ≠ real-world self-paced learning

Model & System Risk

- Persona drift observed in 4 / 6 (Khrushchev, Cao Cao, Ronglu, Titanic crew)
- 5–15 s response latency disrupted attention
- Reproducibility: cached LLM responses + 25 unit tests

Ethics & Privacy

- Informed consent obtained (logs, audio, screen)
- PII removed during transcription
- Compensation: \$10 / hr
- Conversations sent to OpenAI API only (no third-party storage)

7. Conclusion & Future Work



TAKEAWAY 1

Orchestration beats better generation.

CP2 already showed LLMs nail subtasks. CP4 shows the missing layer was a learner-in-the-loop workflow — and once it's there, agency and engagement separate sharply from a static IDN baseline.

TAKEAWAY 2

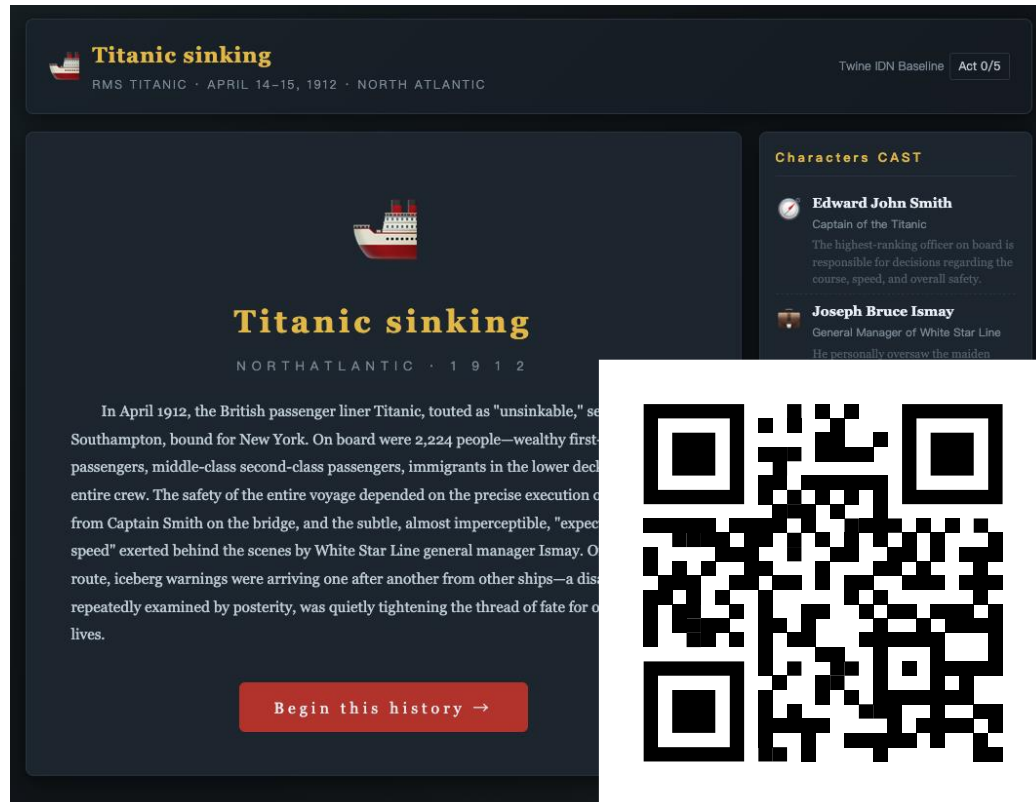
Reading is the bottleneck — visualization is the lever.

Text fatigue was the most consistent friction across all six participants. Multi-voice dialogue is what makes the orchestration feel alive, but it's also why people get tired. The clearest leap forward is visual — portraits, scenes, consequence panels — and ultimately embodied XR, where dialogue happens in space rather than on a wall of words.

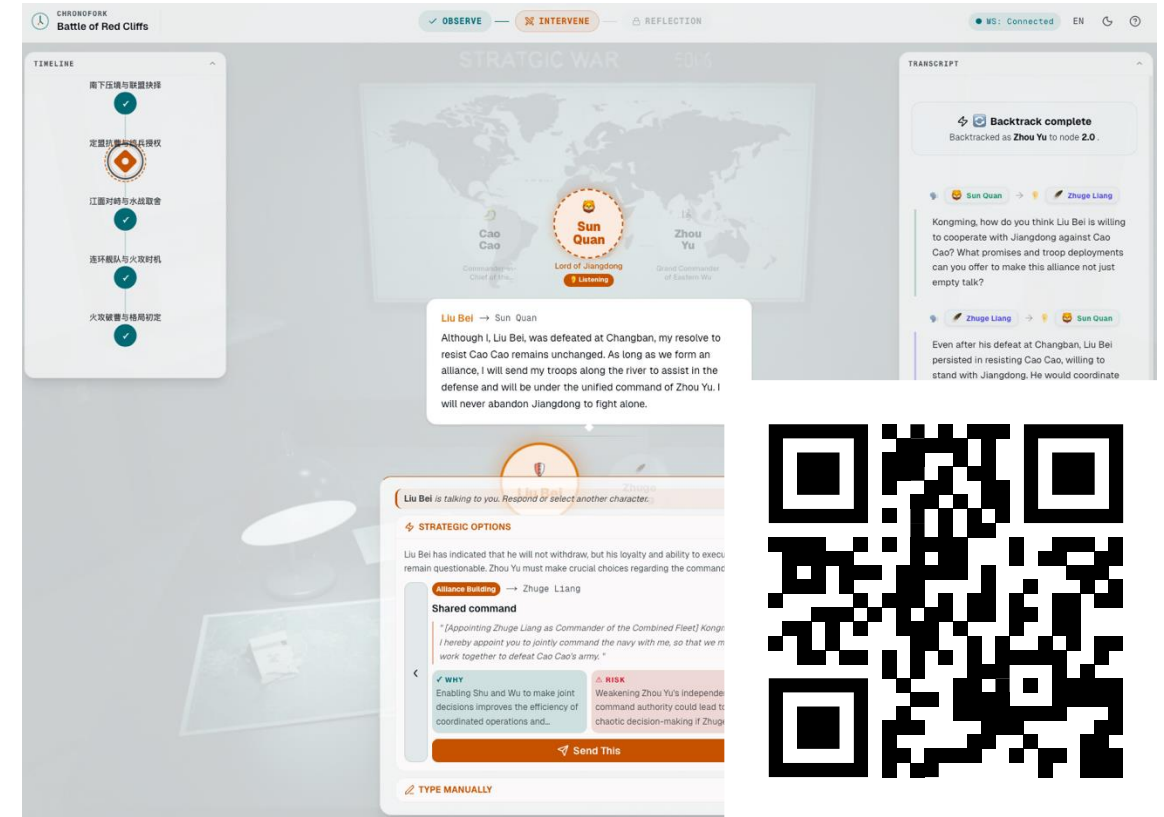
FUTURE WORK

- ▶ **Persona stability layer**
Skill-tagged actors (P3); knowledge-state grounding; persona-break detection
- ▶ **Reflection / outcome panel**
Explicit canonical-vs-counterfactual diff (P3, P6)
- ▶ **Latency & onboarding**
Streaming, "where am I" cues, 30-s walkthrough
- ▶ **Larger evaluation**
20+ participants; retention test 1 week post-study
- ▶ **Immersive thread**
Quest 3 prototype reactivated once web is stable

QA & Try out



Baseline (IDN)
(twine.chronofork.me)



ChronoFork Web
(app.chronofork.me)

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